
Understanding the QIZ Heterogeneous-Firm Trade Model

A Personal Study Guide

Two Regions · Multi-Sector · Endogenous Compliance & Upgrading

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This document explains the QIZ heterogeneous-firm trade model in full detail. It covers the economic theory behind each block — demand, firm costs, trade, compliance, upgrading — and ties each concept to the Python implementation. All key derivations (unit costs, price indices, markups) are worked through step by step. A final section documents the bugs found in the code and the corrections applied.

1. Overview

1.1 What is this model trying to do?

The model asks: **what happens to Egyptian firms and workers when Egypt gains preferential access to the US market through Qualifying Industrial Zones (QIZs)?**

QIZs are special industrial zones in Egypt (and Jordan) where firms can export to the United States **tariff-free**, provided their products contain a minimum share of Israeli inputs. This is the **Rules of Origin (ROO)** requirement. The model captures three central firm-level responses:

- Which firms choose to **comply** with the ROO (and thus export tariff-free to the US).
- Which firms **upgrade their productivity** because they gain US market access.
- How **wages, employment, and welfare** in Egypt change in equilibrium.

1.2 The three-layer structure

The model is solved as three nested loops:

```
OUTER LOOP (General Equilibrium) Solves for: wages {w_Q, w_N} and entry masses {M_rs}
GOODS BLOCK (Price indices & expenditure) Solves for: domestic sector price indices {P_EG,s}
Computes: labor allocation, Egyptian income, sector expenditures, price indices
FIRM PROBLEM (innermost) For every (phi, eps) draw: choose - Which markets to serve (EG, US, RW) - Whether to comply with ROO (Q firms only) - Whether to upgrade productivity
```

1.3 Regions, sectors, and destinations

Symbol	Meaning
Q	QIZ-eligible region
N	Non-QIZ region
T	Textiles / Apparel sector
O	Other manufacturing sector
EG	Domestic Egypt market
US	United States market
RW	Rest of World market

Egypt is modeled as a **small open economy**: it takes foreign prices and expenditures as given. What happens in Egypt does not move US or RW price indices.

2. Demand and Price Indices

2.1 The consumer's problem — nested CES

Consumers in Egypt (and in the US and RW) have **nested CES preferences** — a two-level structure. The upper tier governs how spending is split across sectors; the lower tier governs substitution across varieties within a sector.

2.2 Upper tier — across-sector substitution

The upper-tier utility aggregator is:

$$U_j = [\sum_s \beta_s^{1/\eta} \cdot U_{js}^{(\eta-1)/\eta}]^{\eta/(\eta-1)}$$

where:

- $\beta_s > 0$ are sector expenditure-share weights ($\sum_s \beta_s = 1$)
- $\eta > 0$ is the elasticity of substitution *across* sectors
- U_{js} is the within-sector utility aggregate (defined below)

Think of this as a basket: the consumer can substitute between 'textiles' and 'other goods' with elasticity η . In the code: $\eta = 2.0$, $\beta_T = 0.45$, $\beta_O = 0.55$.

2.3 Lower tier — within-sector (Dixit-Stiglitz)

Within each sector s , consumers value variety (Dixit-Stiglitz aggregator):

$$U_{js} = [\int_{\omega \in \Omega_{js}} q_{js}(\omega)^{(\sigma_s-1)/\sigma_s} d\omega]^{\sigma_s/(\sigma_s-1)}$$

Here $\sigma_s > 1$ is the **elasticity of substitution between varieties** within sector s . A higher σ_s means varieties are closer substitutes — consumers are more price-sensitive. In the code: $\sigma_T = 6.7$, $\sigma_O = 4.0$.

2.4 Deriving individual variety demand

To find how much of variety ω is demanded, maximize the lower-tier utility subject to a sector budget E_{js} . The first-order condition gives:

$$q_{js}(\omega) = (p_{js}(\omega) / P_{js})^{-(\sigma_s)} \cdot (E_{js} / P_{js})$$

Intuition: demand falls when ω 's own price rises (own-price elasticity $-\sigma_s$) and falls when the sector price index P_{js} falls (other varieties become cheaper).

2.5 Deriving the CES price index

The price index P_{js} is defined as the **minimum expenditure needed to attain one unit of sector- s utility**. Substituting optimal demands back into the budget constraint and simplifying yields:

$$P_{js} = \left[\int_{\{\omega \in \Omega_{js}\}} p_{js}(\omega)^{1-\sigma_s} d\omega \right]^{1/(1-\sigma_s)}$$

Key properties:

- Since $1 - \sigma_s < 0$, each variety's contribution $p^{1-\sigma_s}$ is *smaller* for higher-priced varieties — expensive varieties contribute less to the index.
- P_{js} *falls* when more varieties enter (more competition lowers the cost of living) or when existing varieties cut prices.

In the code, the price index is built numerically by summing $p^{1-\sigma}$ contributions across all active firms, then raising to the power $1/(1-\sigma)$:

```
# Accumulate p^{1-sigma} from each active firm contrib += wt * (pEG ** (1.0 - sigma)) # Then convert: P = (sum of p^{1-sigma})^{1/(1-sigma)} P_new[s] = ces_price_from_power(P_pow, sigma)
```

2.6 Revenue of a variety

Revenue = price × quantity:

$$R_{js}(\omega) = E_{js} \cdot (p_{js}(\omega) / P_{js})^{1-\sigma_s}$$

Since $1 - \sigma_s < 0$, a **higher price** → **lower revenue share**. This is the standard CES demand system.

2.7 Aggregate price index across sectors

$$P_{EG} = \left[\sum_s \beta_s \cdot P_{\{EG,s\}}^{1-\eta} \right]^{1/(1-\eta)}$$

The special case $\eta = 1$ (Cobb-Douglas across sectors) is handled separately using the log formula: $P = \prod_s P_s^{\beta_s}$.

2.8 Egyptian income and sector expenditures

Egyptian income:

$$Y_{EG} = w_Q \cdot L_Q + w_N \cdot L_N + T$$

where T is a net transfer that closes the small-open economy (set to zero in the baseline). Upper-tier CES demand gives sector expenditures:

$$E_{\{EG,s\}} = \beta_s \cdot (P_{\{EG,s\}} / P_{EG})^{1-\eta} \cdot Y_{EG}$$

Intuition: if sector s gets relatively cheaper (its price index falls), the expenditure share on that sector rises — when $\eta > 1$, sectors are substitutes.

3. Firm Technology and Unit Costs

3.1 Production function

Each firm produces using labor l and an intermediate input bundle m :

$$y = \phi \cdot l^{\alpha_s} \cdot m^{1-\alpha_s}$$

where:

- ϕ is the firm's **productivity** (higher = more output per input)
- $\alpha_s \in (0,1)$ is the labor share in production
- $1 - \alpha_s$ is the intermediate input share

In the code: $\alpha_T = 0.35$, $\alpha_O = 0.30$.

3.2 Deriving marginal cost — step by step

The firm minimizes total cost $w_r \cdot l + c_m \cdot m$ subject to producing y units.

Step 1 — input demand ratio. The first-order conditions give:

$$l/m = (\alpha_s / (1-\alpha_s)) \cdot (c_m / w_r)$$

Step 2 — solve for inputs in terms of output. Substituting back into the production function:

$$l = (y/\phi) \cdot (\alpha_s / (1-\alpha_s))^{1-\alpha_s} \cdot (c_m / w_r)^{1-\alpha_s}$$

$$m = (y/\phi) \cdot (\alpha_s / (1-\alpha_s))^{-\alpha_s} \cdot (c_m / w_r)^{-\alpha_s} \text{ [after simplification]}$$

Step 3 — total cost. Substituting both into $w_r \cdot l + c_m \cdot m$:

$$C(y) = (y/\phi) \cdot [w_r^{\alpha_s} \cdot c_m^{1-\alpha_s}] / [\alpha_s^{\alpha_s} \cdot (1-\alpha_s)^{1-\alpha_s}]$$

The denominator $\alpha_s^{\alpha_s} \cdot (1-\alpha_s)^{1-\alpha_s}$ is a constant that comes from the algebra of Cobb-Douglas cost minimization.

Step 4 — marginal cost. Since cost is linear in output, $MC = AC$:

$$mc(\phi; c_m) = (1/\phi) \cdot w_r^{\alpha_s} \cdot c_m^{1-\alpha_s} / [\alpha_s^{\alpha_s} \cdot (1-\alpha_s)^{1-\alpha_s}]$$

In the code:

```
alpha = p['alpha'][s] denom = (alpha ** alpha) * ((1.0 - alpha) ** (1.0 -
alpha)) mc_base = (1.0 / (phi * delta)) * (w_r**alpha * cN**(1-alpha)) /
denom
```

Here $cN = p_{rw[s]}$ is the freely-sourced intermediate price and δ is the upgrading multiplier (1 if no upgrade). Notice that $mc \propto 1/\phi$: more productive firms have lower marginal cost.

3.3 Markup pricing — derivation

Under monopolistic competition with CES demand, every firm charges a **constant markup** over marginal cost. To see why, the firm faces demand $q = A \cdot p^{-(1-\sigma_s)}$ where A collects all terms outside the firm's control. Revenue is $R = p \cdot q = A \cdot p^{1-\sigma_s}$. Profit is:

$$\pi = R - mc \cdot q = A \cdot [p^{1-\sigma_s} - mc \cdot p^{-\sigma_s}]$$

Setting $d\pi/dp = 0$:

$$A(1-\sigma_s) p^{-\sigma_s} + A \sigma_s \cdot mc \cdot p^{-(\sigma_s-1)} = 0$$

Solving for p :

$$p = \sigma_s / (\sigma_s - 1) \cdot mc = \mu_s \cdot mc$$

The **markup** $\mu_s = \sigma_s / (\sigma_s - 1)$ is higher when σ_s is lower (less substitutable varieties → more market power):

- Textiles: $\sigma_T = 6.7 \rightarrow \mu_T = 6.7/5.7 \approx 1.175$ (17.5% markup)
- Other mfg: $\sigma_O = 4.0 \rightarrow \mu_O = 4/3 \approx 1.333$ (33.3% markup)

3.4 Variable profit

A key CES identity: for any destination j , variable profit is always a fixed fraction of revenue:

$$\pi^{var}_{rjs} = R_{rjs} - mc \cdot q_{rjs} = R_{rjs} - R_{rjs}/\mu_s = R_{rjs} / \sigma_s$$

This is why the code uses $\pi_{EG} = R_{EG} / \sigma$, $\pi_{US} = R_{US} / \sigma$, etc.

4. Trade Costs, Tariffs, and Pricing

4.1 Iceberg trade costs

Shipping goods to destination j from region r incurs an **iceberg cost** $d_{rjs} \geq 1$: to deliver 1 unit, the firm must ship d_{rjs} units. The extra $d_{rjs} - 1$ units 'melt' in transit. This scales the effective marginal cost of delivery to:

$$\text{Effective delivery cost} = d_{rjs} \cdot mc_{rs}(\phi)$$

In the baseline: $d_{Q,US,s} = 1.35$, $d_{N,US,s} = 1.50$ (QIZ firms have lower US trade costs), $d_{\cdot,EG,s} = 1.0$ (no domestic shipping cost).

4.2 MFN tariff for US exports

Non-compliant firms exporting to the US face a **Most Favored Nation (MFN) tariff** t^{MFN}_s on top of the iceberg cost. The total delivered wedge is:

$$\tau_{rjs} = d_{rjs} \times \begin{cases} 1 + t^{MFN}_s & \text{if } j=US \text{ and non-compliant} \\ 1 & \text{if } j=US \text{ and compliant (QIZ)} \\ 1 & \text{if } j \in \{EG, RW\} \end{cases}$$

Compliant QIZ firms pay **zero tariff** — that is the core benefit of the QIZ arrangement. In the baseline: $t^{MFN}_T = 0.15$, $t^{MFN}_O = 0.03$.

4.3 Delivered price

The firm sets price at markup over delivered marginal cost:

$$p_{rjs}(\phi) = \mu_s \cdot \tau_{rjs} \cdot mc_{rs}(\phi)$$

In the code:

```
p_EG = mu * (tau_EG * mc_base) p_RW = mu * (tau_RW * mc_base) p_US = mu *  
(tau_US * mc_US) # mc_US differs for compliant firms
```

Note: mc_{US} can differ from mc_{base} for compliant firms, because compliance changes the intermediate input used for US-destined production (see Section 6).

5. Export Participation

5.1 The fixed cost condition

Serving each destination requires paying a **fixed cost** $w_r \cdot f_{rjs}$ (in units of labor). A firm serves destination j if and only if variable profit covers the fixed cost:

$$R_{rjs}(\phi) / \sigma_s \geq w_r \cdot f_{rjs}$$

Since $R_{rjs} \propto \phi^{(\sigma_s-1)}$ (more productive firms charge lower prices and capture higher revenue shares), there exists a **productivity cutoff** ϕ_{rjs}^* such that only firms with $\phi \geq \phi_{rjs}^*$ find it profitable to serve market j . In the code:

```
serve_EG = (pi_EG >= w_r * f_dom) serve_US = (pi_US >= w_r * f_US) serve_RW =  
(pi_RW >= w_r * f_RW)
```

5.2 Market hierarchy

In practice, the domestic market (EG) is the easiest to serve: it has the lowest fixed cost and no iceberg losses. Exporting to US or RW requires additional fixed costs and trade costs. This generates a natural hierarchy:

- All exporters also serve the domestic market.
- Not all domestic sellers export.
- US exporters face a higher fixed cost than RW exporters (in the baseline).

5.3 Pareto productivity distribution

Each entrant in (r, s) draws productivity ϕ from a Pareto distribution:

$$G_s(\phi) = 1 - (\phi_{min,s} / \phi)^{\theta_s}, \phi \geq \phi_{min,s}$$

The shape parameter θ_s must satisfy $\theta_s > \sigma_s - 1$ to ensure finite expected firm size. Higher θ_s concentrates more firms near the minimum — a fatter left tail of low-productivity firms.

In the code, the Pareto distribution is approximated numerically using a **midpoint quadrature rule**:

```
def pareto_grid(phi_min, theta, n): u = (np.arange(n) + 0.5) / n # uniform  
draws in (0,1) phi = phi_min * (1.0-u)**(-1.0/theta) # inverse Pareto CDF w =  
np.full(n, 1.0/n) # equal probability weights return phi, w
```

The u values are evenly spaced interior points; the inverse CDF maps them to the Pareto support; weights are uniform because the grid is uniform over probability mass.

6. Rules of Origin (ROO) Compliance

6.1 What is the ROO?

The QIZ agreement grants tariff-free access to the US market on the condition that the exported product contains at least a fraction γ_s of Israeli inputs. This creates a **trade-off**:

- **Benefit:** zero US tariff instead of t^{MFN}_s .
- **Cost:** must use more expensive Israeli inputs, plus administrative burden.

Only firms in the **QIZ-eligible region** ($r = Q$) can comply. Non-QIZ firms ($r = N$) cannot access the preferential rate regardless.

6.2 The intermediate input cost under compliance

A compliant firm must use an input bundle with fraction γ_s Israeli content. This is modeled as a **normalized Cobb-Douglas** quantity index:

$$m = m_{IL}^{\gamma_s} \cdot m_{RW}^{(1-\gamma_s)} / [\gamma_s^{\gamma_s} \cdot (1-\gamma_s)^{(1-\gamma_s)}]$$

The denominator $K = \gamma_s^{\gamma_s} \cdot (1-\gamma_s)^{(1-\gamma_s)}$ is the **normalization constant**. This is essential.

6.3 Why normalization matters

Without normalization, switching from free-market sourcing to any Cobb-Douglas mix would mechanically inflate unit cost — even if Israeli and world inputs cost exactly the same. The normalization ensures that if $p_{IL} = p_{RW}$, compliance has **zero marginal cost in unit cost terms**.

The ROO only bites when Israeli inputs are genuinely more expensive than world inputs.

6.4 Deriving the compliant unit cost

The unit cost of the normalized input bundle is found by cost minimization over (m_{IL}, m_{RW}) . The normalization constant in the quantity index cancels in the derivation, leaving:

$$c^{mix}_{\{m,s\}}(\gamma_s) = p_{IL}^{\gamma_s} \cdot p_{RW}^{(1-\gamma_s)}$$

This is a weighted geometric mean of input prices.

Verification: if $p_{IL} = p_{RW} = p$, then $c^{mix} = p^{\gamma_s} \cdot p^{(1-\gamma_s)} = p$. Cost equals the common input price — no compliance penalty. ✓

In the code:

```
def cmix_normalized(p_il, p_rw, gamma): return (p_il ** gamma) * (p_rw ** (1.0 - gamma))
```

6.5 Administrative wedge

Beyond input costs, compliance creates paperwork and coordination costs that increase with the stringency γ_s :

$$\chi_s(\gamma_s) = 1 + \xi_s \cdot \gamma_s, \xi_s > 0$$

The total compliant intermediate cost for US-destined shipments is:

$$c^{C,US}_{\{m,s\}} = \chi_s(\gamma_s) \cdot c^{mix}_{\{m,s\}}(\gamma_s)$$

And the compliant marginal cost for US production:

$$mc_{C_US} = (1/\phi) \cdot w_r^{\alpha_s} \cdot (c^{C,US}_{\{m,s\}})^{(1-\alpha_s)} / [\alpha_s^{\alpha_s} \cdot (1-\alpha_s)^{(1-\alpha_s)}]$$

In the baseline: $\xi_T = 1.0$, $\xi_O = 0.6$, $\gamma_T = \gamma_O = 0.105$.

6.6 ROO applies only to US shipments

This is a deliberate and important modeling choice. Non-US production (domestic EG, RW exports) uses the free-market input $c^N_m = p_{RW,s}$. Only US-destined output must satisfy the Israeli content requirement. This is consistent with how QIZ certification works in practice: certification is at the **shipment level**, not the firm level.

6.7 Partial compliance — idiosyncratic fixed costs

Even among firms that benefit from compliance in expectation, not all comply. Each firm draws an idiosyncratic compliance fixed cost:

$$f^C_{\{i,s\}} = f^C_s \cdot \exp(\varepsilon_i), \varepsilon_i \sim N(0, \sigma^2_{\{C,s\}})$$

So $f^C_{\{i,s\}}$ is **lognormally distributed**: the median compliance cost is f^C_s , but some firms face much higher costs (e.g., their supply chain is hard to restructure). This generates **partial take-up**: some firms comply, some don't, even among those serving the US.

*Note: f^C_s (fC_mean in the code) is the **median**, not the mean, of the lognormal. The true mean is $fC_mean \cdot \exp(\sigma^2_{\{C,s\}} / 2)$. For textiles: $mean \approx 0.35 \cdot \exp(0.125) \approx 0.397$. This matters if calibrating to survey data on average compliance expenditure.*

6.8 The compliance decision in the code

For Q-region firms, the code enumerates **four strategy combinations**:

comply	upgrade	Description
False	False	Baseline: no compliance, no upgrade
False	True	Upgrade only (if US is served)
True	False	Comply with ROO, no upgrade
True	True	Comply and upgrade

For each strategy, the code computes total profit and picks the best. Compliance is only effective if the firm actually serves the US — compliance without US access is a dominated strategy and is skipped.

7. Productivity Upgrading

7.1 The mechanism

Serving the US market exposes firms to demanding buyers, quality standards, and global value chains — this triggers **productivity upgrading**. Any firm that serves the US can pay a fixed cost $w_r \cdot f^U_s$ to permanently scale its productivity:

$$\phi' = \delta_s \cdot \phi, \delta_s > 1$$

In the baseline: $\delta_T = 1.12$, $\delta_O = 1.07$, $f^U_T = 1.8$, $f^U_O = 2.3$ (labor units).

7.2 Why upgrading helps across all markets

Under CES demand with markup pricing, revenue in any destination j satisfies:

$$R_{\{rjs\}} \propto \phi^{(\sigma_s - 1)}$$

This is because lower $mc \propto 1/\phi$ translates to a lower price, which expands demand. After upgrading, $\phi' = \delta_s \cdot \phi$, so:

$$R'_{\{rjs\}} / R_{\{rjs\}} = \delta_s^{(\sigma_s - 1)}$$

This ratio is **identical across all destinations**. Upgrading lifts revenues in Egypt, the US, *and* the Rest of World simultaneously. This is the key **spillover mechanism**: firms that upgrade because of US access also become more competitive in third markets.

7.3 The US-access constraint

Upgrading is only available if the firm serves the US (parameter *upgrade_requires_US* = *True*). In the code:

```
if upgrade and p['upgrade_requires_US'] and (not serve_US): eff_upgrade =  
False # cannot upgrade without US market access
```

If a firm plans to upgrade but does not end up serving the US (because the profit threshold is not met), the upgrade is dropped and profits are recomputed without it. This reflects the economic story: upgrading is triggered by learning-by-exporting to the US.

8. General Equilibrium

8.1 Labor mobility

Workers choose between regions Q and N based on real wages. The **logit** allocation is:

$$\lambda_r = (w_r / P_{EG})^\kappa / \sum_{r'} (w_{r'} / P_{EG})^\kappa$$

$$L_r = \lambda_r \cdot L$$

Higher κ means workers are more responsive to wage differences (more mobile). At $\kappa \rightarrow \infty$ wages equalize. At $\kappa = 0$ labor is immobile. Baseline: $\kappa = 2.0$.

8.2 Free entry

In each (r, s) , firms pay entry cost $w_r \cdot f^E_{rs}$ to draw a productivity. In equilibrium, the **expected profit from entry equals the entry cost**:

$$E[\Pi_{rs}(\phi, \varepsilon)] = w_r \cdot f^E_{rs}$$

This pins down the mass of entrants M_{rs} . If expected profits exceed entry cost, more firms enter, driving down profits (via the price index) until equality holds.

8.3 Labor market clearing

In each region, total labor demand must equal labor supply:

$$L_r = \sum_s M_{rs} \cdot [E[l^{\text{var}}_{rs}] + E[l^{\text{fix}}_{rs}] + f^E_{rs}]$$

where:

- $E[l^{\text{var}}_{rs}]$ = expected variable labor from production
- $E[l^{\text{fix}}_{rs}]$ = expected fixed labor (domestic, export, compliance, upgrade costs)
- f^E_{rs} = entry cost per entrant

8.4 The outer loop solver

The solver updates wages and entry masses using a log-linear tatonnement:

```
# If labor demand > supply: raise wage w[r] *= exp(step * log(Ld[r] / Ls[r]))
# If E[profit] > w*f_E: too profitable, more firms enter M[r,s] *= exp(step *
log(E_profit / entry_cost))
```

This adjusts wages up when labor demand exceeds supply, and entry masses up when profits exceed entry costs. The step size `outer_step = 0.10` controls the speed (smaller = more stable, slower).

8.5 Welfare

Welfare is real income — how much the composite good Egyptian households can afford:

$$W = Y_{EG} / P_{EG}$$

9. Counterfactuals

9.1 Counterfactual A — Shut down the productivity channel

Set $\delta_s = 1$ for all sectors (no upgrading benefit). Re-solve the full general equilibrium. Compare with baseline across:

- **Compliance rates:** without upgrading, the only benefit of compliance is tariff-free US access. Compliance rates may fall.
- **RW exports:** without the upgrading spillover, compliant firms won't gain RW competitiveness. This tests whether upgrading drives the observed pattern.
- **Wages and welfare:** if upgrading is important, shutting it down reduces wages in Q and aggregate welfare.

In the code:

```
p_off['delta'] = {s: 1.0 for s in p['sectors']}
```

9.2 Counterfactual B — Vary the ROO content requirement γ_T

Solve GE for a grid of $\gamma_T \in [0, \gamma_{\max}]$, holding everything else fixed. For each γ_T , record: welfare, wages, compliance rates, US exports, RW exports, and implied Israeli input intensity.

What to expect along the γ_T path:

- $\gamma_T = 0$: no Israeli content required. Compliance is cheap (no input cost change, only paperwork). High compliance rate.
- γ_T increases: compliance becomes more costly. Fewer firms comply. But those that do use more Israeli content.
- **Welfare $W(\gamma_T)$:** traces the policy frontier — the trade-off between Israeli conditionality and Egyptian welfare gains from tariff-free US access.

The baseline uses $\gamma_T = \gamma_O = 0.105$ (10.5% Israeli content requirement).

10. Bugs Found and Corrections Applied

The following issues were identified in *qiz_model_ge.py*. All have been corrected in the source file.

Bug 1 — `serve_EG` not rechecked after upgrade fallback [HIGH]

Location: `firm_best()`, lines ~286–321

What happened: When a firm plans to upgrade (`upgrade=True`) but fails to serve the US — so the upgrade is dropped — the code recomputes `R_EG` under the non-upgrade marginal cost. However, the flag `serve_EG` was never updated. A firm only marginally active in Egypt *because* of the upgrade boost was incorrectly recorded as active after the upgrade was dropped.

Economic consequence: Corrupts the domestic price index (a non-viable variety enters it), overstates labor demand in Egypt, and inflates free-entry expected profits.

Fix applied: After the upgrade fallback block, `serve_EG` is now rechecked: if $R_EG/\sigma < w_r * f_dom$ after losing the upgrade, the firm is set to fully inactive.

Bug 2 — `normal_grid` uses endpoint-inclusive `linspace` [MEDIUM]

Location: `normal_grid()`, lines ~154–162

What happened: The docstring claims 'midpoint-rule quadrature' but `np.linspace(-a, a, n)` includes the endpoints $-a$ and $+a$. A true midpoint rule uses only interior points. The step size $h = 2a/(n-1)$ was also the `linspace` spacing, not the midpoint spacing $2a/n$.

Fix applied: Replaced `linspace` with the midpoint formula:

$x = -a + (\text{arange}(n) + 0.5) \times (2a/n)$, $h = 2a/n$

Bug 3 — `print_key` references `p` which is not in scope [HIGH]

Location: `print_key()`, line ~637

What happened: The function `print_key(sol, label)` used `p['L_total']` but `p` was not a parameter. It worked only because `p` was defined globally in `__main__`. If called from any other context, it raises `NameError`.

Fix applied: Added `p` as an explicit third parameter: `print_key(sol, label, p)`.

Bug 4 — A line in print_key always printed 1.0 [MEDIUM]

Location: print_key(), line ~636

What happened: The expression `sol['Ls'][r] / sol['goods']['Ls'][r]` always equals 1.0 because `sol['Ls']` and `sol['goods']['Ls']` are the *same Python object* (both reference the same dict). The comment even admitted 'placeholder'. This printed `{'Q': 1.0, 'N': 1.0}` every time, providing no information.

Fix applied: Removed the redundant line entirely.

Bug 5 — Compliance fallback used $-1e18$ sentinel [MEDIUM]

Location: firm_best(), lines ~328–337

What happened: When `comply=True` but `serve_US=False`, the code set `profit = -1e18` to ensure this dominated strategy never wins. This is fragile: it relies on (`comply=False`, `upgrade=False`) always being in the strategy set. Any future refactoring that reorders or removes strategies could silently break this.

Fix applied: Replaced the sentinel with a clean 'continue' to skip the dominated strategy entirely.

Bug 6 — Shallow copy in counterfactual [LOW]

Location: counterfactual_shutdown_productivity(), line ~600

What happened: `p_off = {k: v for k,v in p.items()}` is a shallow copy. It is currently safe because `p_off['delta']` is immediately reassigned to a new dict. But in-place mutation of any nested dict would silently corrupt the original `p`.

Fix applied: Replaced with `copy.deepcopy(p)`.

Calibration Note — `fC_mean` is the median, not the mean

Location: params_defensible(), lines ~109–110

The compliance cost $fC_i = fC_mean \times \exp(\sigma_C \times \epsilon)$ defines a lognormal where `fC_mean` is the **median**. The **mean** is $fC_mean \times \exp(\sigma_C^2 / 2)$. For textiles: $mean \approx 0.35 \times \exp(0.125) \approx 0.397$ (13% above the median). If calibrating to average compliance expenditure from survey data, the calibration target should be the mean, not `fC_mean`.

Modeling Note — No explicit wage numeraire

Location: `solve_equilibrium()`, line ~521

No wage is held fixed during iteration. The model is implicitly pinned by the exogenous foreign price indices $P_{\text{foreign}} = 1.0$. If foreign prices are ever changed, the absolute wage level becomes unanchored. Best practice: after each update step, normalize one wage (e.g., $w_Q = 1.0$) and rescale accordingly.

All six bugs have been corrected in the source file `qiz_model_ge.py`. The two notes are flagged for calibration awareness but do not require code changes for the current baseline.